

Module 09: Intellectual Property Rights (IPR) and Technology Transfer

Intellectual Property Rights:

Intellectual Property Rights (IPR) are legal rights that protect people's ideas and creations. They cover things like inventions, art, brand names, logos, pictures, and designs used in business. These rights help creators get credit or earn money from what they make.

Types of IPR:

Type	Description	Example
Patent	Protects inventions (products or processes).	New drug formula.
Copyright	Protects original works of authorship.	Books, music, software.
Trademark	Protects brand names, slogans, logos.	Nike "swoosh", CocaCola.
Design Rights	Protects the visual design of objects.	Shape of a bottle.
Geographical Indication (GI)	Protects products tied to a location.	Darjeeling tea, Basmati rice.
Trade Secret	Protects confidential business information.	Coca-Cola recipe.

The four main types of IPR are:

1. Patents

Definition:

A patent is a legal right (P) granted to an inventor for a limited period of time (usually 20 years) in exchange for publicly disclosing the invention.

Key Features:

- Gives the inventor exclusive rights for 20 years (in most countries).
- The invention must be new, not obvious, and useful.
- The inventor must publicly share details about how it works.

Example:

A battery that charges in 5 minutes and works for 10 days. If this battery is truly new and useful, it can get a patent.

Example in Real Life:

The **EpiPen** (a device for injecting allergy medicine) is protected by patents so that others can't copy the same idea.

2. Trademarks

Definition:

A **trademark** is a word, sign, logo, or symbol that helps people know which company's product or service it is, and sets it apart from others.

Key Features:

- Builds a strong brand identity.
- Can last forever if the trademark is still used.
- If someone copies it, it can confuse customers and cause legal problems.

Example:

The Nike “**swoosh**” logo is a trademark — it shows the product is made by Nike.

Example in Real Life:

Coca-Cola — the name, logo, and even the unique bottle shape are trademarked.

3. Copyrights

Definition:

A **copyright** protects original works like books, songs, movies, software, and artwork from being copied or used without permission.

Key Features:

- It is automatic once the work is created (registration is not required but can help).
- Lasts for the creator's lifetime plus 60 years (in India).
- Protects how the idea is shown, not the idea itself.

Example:

A novel, a piece of music, or software code. Others can't copy or sell it without the creator's permission.

Example in Real Life:

The **Harry Potter** books are copyrighted — only J.K. Rowling and her publishers can publish or sell them.

4. Trade Secrets

Definition:

A trade secret is a secret method, formula, or process that gives a company an advantage over others and is not shared with the public.

Key Features:

- Kept secret and not made public.
- No official registration needed.
- Stays protected as long as it stays secret.

Example:

The **Coca-Cola** drink formula — it has stayed secret for over 100 years.

Example in Real Life:

Google's **search algorithm** is a trade secret that helps it stay ahead of other companies.

Importance of IPR

- Encourages people to invent and create new things.
- Helps creators and businesses make money by selling their ideas.
- Protects the money and effort spent on research and development (R&D).
- Promotes healthy competition and builds trust with customers.

1. Patenting Process

The **patenting process** is a set of steps to legally protect a new invention. A patent gives the inventor the exclusive right to make, use, and sell the invention for a limited time (usually 20 years).

Steps in the Patenting Process:

Step 1: Idea/Concept Development

- Create a new and useful invention.
- Make sure it is something new, not obvious, and can be made and used in industry.

Example: A company invents a solar-powered water purifier with Nano filters.

Step 2: Patent Search

- Check if anyone has already patented the same idea.
- Use tools like Google Patents, USPTO, WIPO databases.

Example: Search worldwide to see if any other solar-powered purifier with Nano filters exists.

Step 3: Prepare Patent Application

- Write the patent document. It should include:
 - o Title
 - o Abstract (short summary)
 - o Background (existing problems)
 - o Detailed Description (how it works)
 - o Claims (what exactly is protected)

Example: Write the details of how the purifier works and what makes it special.

Step 4: File the Patent Application

- Submit the application nationally or internationally:
 - o **Provisional Application (optional):** Gives 12 months to finish and file the complete version.
 - o **Non-Provisional (Complete) Application:** Has all details for the patent office to review.

Example: File it in India's patent office (IPO) and under PCT (Patent Cooperation Treaty) for global protection.

Step 5: Publication

- The patent application is published 18 months after filing.
- Anyone can read it and give feedback or objections.

Step 6: Examination

- Request the patent office to examine the application (called Request for Examination – RFE).
- The office checks if the invention really is new and useful, and may ask questions.

Example: An examiner questions if the special filter is truly new — the inventor explains with proof.

Step 7: Grant of Patent

- If the invention passes all checks, the patent is approved and listed in the patent journal.

Step 8: Maintenance

- The inventor must pay annual fees to keep the patent active.

2. IP Protection Strategies

Organizations use different ways to protect their intellectual property (IP) like patents, copyrights, trademarks, and trade secrets.

Types of IP Protection Strategies:

A. Patents

- Protect inventions and new technologies.
- **Strategy:** File patents in important countries; allow other companies to use the patents through licenses.

Example: Tesla patents its electric car technology but also gives free licenses to promote electric cars.

B. Trademarks

- Protect the brand's name, logo, slogan, or product look.
- **Strategy:** Register trademarks in all countries where the company sells its products to stop misuse.

Example: Coca-Cola protects its red logo and special font all over the world.

C. Copyrights

- Protect creative works like books, music, movies, art, or software.
- **Strategy:** Use tools like DRM (Digital Rights Management), watermarks, and official registration to stop copying.

Example: Microsoft copyrights Windows and uses product keys to prevent illegal copies.

D. Trade Secrets

- Protect secret business information like formulas, algorithms, or strategies.
- **Strategy:** Use NDAs (Non-Disclosure Agreements), limit who can access the secret, and use encryption.

Example: Coca-Cola keeps its drink formula a secret instead of patenting it.

E. Licensing and Cross-Licensing

- Allow others to use your IP in exchange for royalty payments, while you keep ownership.

Example: Qualcomm licenses its mobile chip patents to phone makers and earns royalties.

F. Defensive Publishing

- Publish an idea to make sure no one else can patent it later.

Example: IBM often publishes technical details so competitors can't patent common tech ideas.

G. Open Innovation / Open Source

- Share some parts of your technology with others to encourage teamwork but keep control over key parts.

Example: Google makes Android open source but keeps Google Play services under its own control.

Technology Transfer:

Technology Transfer refers to the process of sharing or disseminating technology between organizations (like universities to industry) or countries.

Modes of Technology Transfer:

Mode	Description
Licensing	Allowing others to use IP for a fee or royalty.
Joint ventures	Collaborative projects sharing tech.
Franchising	Providing a business model and technology.
Spin-offs/startups	New companies formed to commercialize tech.
Consultancy and technical services	Expertise sharing.

Relationship between IPR and Technology Transfer:

- **IPR helps in valuing technology** — stronger IP means higher value.
- **IPR provides legal security** during transfer. Licensing of IP is a **major mechanism** for technology transfer.
- Universities and R&D institutions **commercialize research** through patenting and then transferring.

Challenges in IPR and Technology Transfer:

- **Lack of awareness** among researchers.
- **High cost** of filing and maintaining IPR.
- **Weak enforcement** of IP laws in some countries.
- **Complex negotiation** in licensing agreements.
- **Cultural and legal differences** in international transfers.

Licensing Agreements and IP Valuation

1. Licensing Agreements

Definition:

A **Licensing Agreement** is a legal contract where the IP owner (**licensor**) gives permission to another party (**licensee**) to use the IP under certain conditions — but the owner keeps the IP.

Key Parts:

- What the license covers (only for one party or many, where, how long)
- How much the licensee pays (royalties or fees)
- What rights are given (copy, sell, change, etc.)
- Keeping information secret
- Rules for ending the agreement

Types of Licensing Agreements:

1. **Patent License:** Lets someone use a patented invention.
2. **Trademark License:** Lets someone use a brand name or logo.
3. **Copyright License:** Lets someone copy or share books, music, art, software, etc.
4. **Trade Secret License:** Gives access to secret business information like formulas or special processes.
5. **Software License:** Lets someone use software under agreed rules.

Example:

Microsoft gives Dell a license to use its **Windows OS**.

- Microsoft is the licensor, Dell is the licensee.
- Dell pays Microsoft a fee for each computer sold with Windows installed.

The contract explains where it applies, how long it lasts, rules for updates, and other limits.

IP Valuation

Definition:

IP Valuation means finding out how much an intellectual property is worth in money.

Why It's Important:

- To set fair license prices.
- When companies merge or are bought.
- To get investments or loans.
- For court cases and damage claims.
- For accounting and tax reporting.

Ways to Value IP:

1. Cost-Based Method

- Based on how much it cost to make or develop the IP.
- Good for new or early-stage inventions.
- **Example:** If a company spent ₹10 lakh to create a software tool, that's its cost-based value.

2. Market-Based Method

- Looks at what similar IP has sold for.
- Needs real market data to compare.
- **Example:** If a similar mobile game license sold for ₹5 crore, that price can be used as a guide.

3. Income-Based Method

- Estimates how much money the IP will make in the future and works out what that is worth today.
- Uses a method called Discounted Cash Flow (DCF).
- **Example:** If a patented drug is expected to earn ₹1 crore every year for 10 years, you calculate its present value using a discount rate.

Combined Example:

A biotech company has a patent for a **COVID-19 test kit**.

- It gives a pharma company permission to use the patent for 5 years and charges ₹10 per kit sold.
- They expect to sell 1 crore kits every year.
- So, the yearly income = $₹10 \times 1 \text{ crore} = ₹10 \text{ crore per year}$.
- When you discount this income over 5 years, the total value of the IP comes to about ₹38 crore.

1. Open Source

Definition:

Open Source refers to software (or any work) whose source code is made freely available to the public. Anyone can view, modify, and distribute the code as long as they comply with the license terms.

Key Characteristics:

- **Transparency:** The source code is visible to all.
- **Collaboration:** Anyone can contribute to its improvement.
- **Freedom:** Users can modify and use it as they wish.
- **Community-driven:** Maintained by communities rather than a single company.

Examples:

a. Linux Operating System

- **Details:** Linux is an open-source OS kernel. Various distributions like Ubuntu, Fedora, and Debian are built around it.
- **Use Case:** Used in servers, desktops, embedded systems, and even Android smartphones.
- **Benefit:** Free to use and customize. Backed by a strong community.

b. Apache Web Server

- **Details:** A widely used open-source web server software.
- **Use Case:** Powers a large portion of the websites on the internet.
- **Benefit:** Secure, flexible, and frequently updated by developers worldwide.

2. Open Innovation

Definition:

Open Innovation is a business management model that promotes collaboration with people and organizations outside the company to drive innovation. Instead of relying solely on internal R&D, firms use external ideas and technologies.

Key Characteristics:

- **External collaboration:** With universities, startups, customers, or even competitors.
- **Shared risks and rewards.**
- **Faster problem-solving and innovation cycles.**

Examples:

a. Procter & Gamble's "Connect + Develop" Initiative

- **Details:** P&G collaborates with external inventors and companies to co-develop new products.
- **Example:** The Swiffer cleaning system was developed in collaboration with an external partner.
- **Benefit:** Accelerated product development and innovation.

b. Lego Ideas Platform

- **Details:** Fans submit new Lego set ideas. If selected, the idea is turned into a real product.
- **Use Case:** Lego —Women of NASA set originated from user suggestion.
- **Benefit:** Engages community, generates fresh product ideas, and shares profit with creators.